Project Design Phase
Solution Architecture

Date	30 September 2026
Team ID	9100Spark
Project Name	ImportData using Transform Maps
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture bridges the business problem (moving bulk employee data from Excel into ServiceNow accurately) and the technology solution (Import Sets and Transform Maps on the ServiceNow platform). Its goals for this project are to:

- Find the right ServiceNow features to load, map and transform spreadsheet data without custom code.
- Describe the structure and behaviour of the import solution to the project stakeholders.
- Define the features, development phases (four milestones) and solution requirements.
- Provide the specification by which the import is defined, managed and validated.

Solution Architecture Diagram:

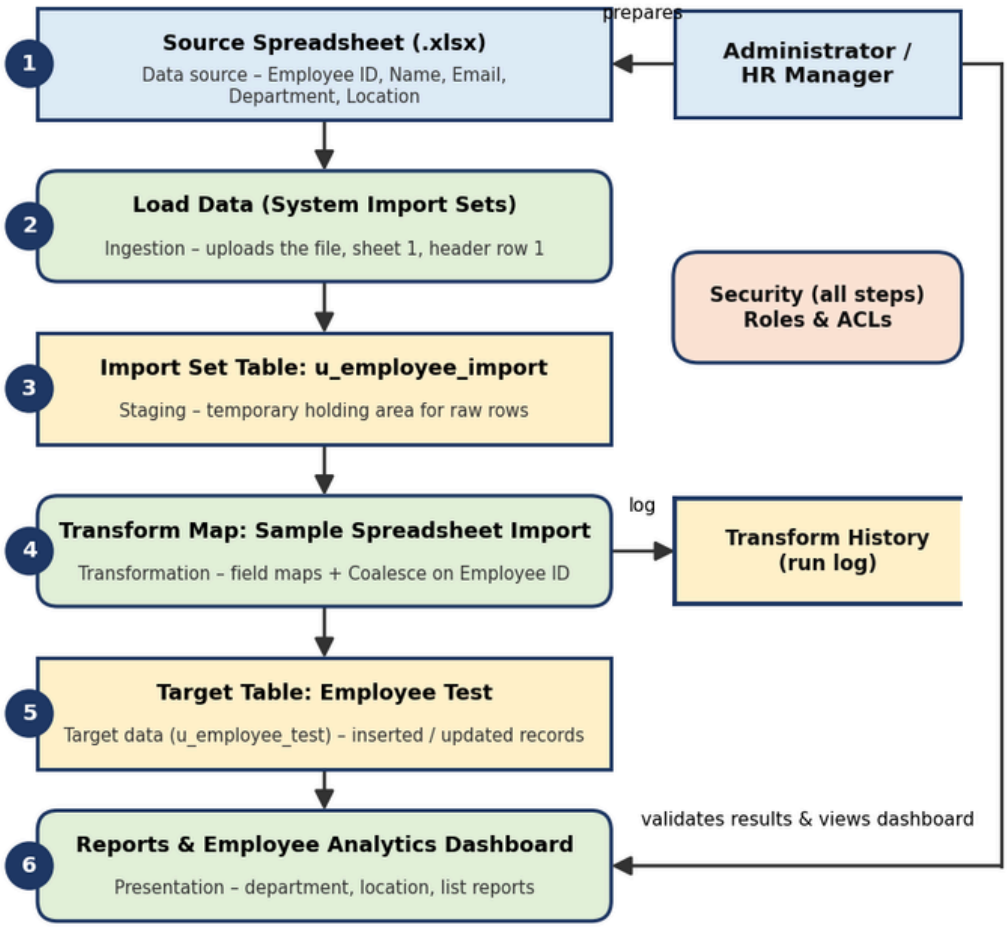


Figure 1: Architectureand data flow of the employee data import solution

Architecture layers and data flow:

Step	Layer	Component	Role in the solution
1	Data source	Source spreadsheet (.xlsx)	Employee ID, Name, Email, Department and Location prepared in Google Sheets and downloaded as Sample Spreadsheet.xlsx
2	Ingestion	Load Data (System Import Sets)	Uploads the file (sheet 1, header row 1) and creates the Import Set table
3	Staging	Import Set table u_employee_import	Temporary holding area for the raw rows before validation and transformation
4	Transformation	Transform Map: Sample Spreadsheet Import	Maps source fields to target fields (Auto Map Matching Fields); Coalesce on Employee ID inserts new and updates existing records
5	Target data	Employee Test table u_employee_test	Final employee records; checked after the transform and Transform History
6	Presentation	Reports and Employee Analytics Dashboard	Employees by Department, Employees by Location and Employee List reports shown together on one dashboard

Step	Layer	Component	Role in the solution
–	Cross-cutting	Roles, ACLs and Transform History	Access control for loading data and dashboards; log of inserted, updated, ignored and skipped rows for every run